

KAHU HUTTON

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About Me

I'm a third-year Electrical Engineering student at the University of Canterbury who enjoys solving challenging problems by integrating hardware and software. I specialize in embedded system design, having played a key role in spacecraft telemetry projects at Dawn Aerospace, and thrive in collaborative, high-pressure environments.

Education

University of Canterbury

Bachelor of Electrical Engineering with Honours

2023 – Present

Christchurch, NZ

Experience

Dawn Aerospace

Nov 2024 – Feb 2025

Spacecraft Propulsion Development Intern

Christchurch, NZ

- Assisted in software and hardware integration for satellite propulsion, ensuring smooth communication between thrusters, sensors, and telemetry modules.
- Played a key role in Feedline Characterization: developed vacuum-compatible test jig and GUI with YAMCS for real-time control, data logging, and thermal-loss assessment.
- Automated life-cycle testing of satellite electronics under heat, vacuum, and vibration by designing igniter test hardware, implementing CAN/RS422 telemetry, and setting up Grafana dashboards with alerts.
- Authored comprehensive documentation (installation guides, user manuals) for in-house tools, streamlining new-hire onboarding.

KORA — Education Platform

Apr 2024 – Present

Founder & Lead Developer

Christchurch, NZ

- Developed a full-stack web app with React, Next.js, Docker, and GCP, funded by a \$50,000 NZD Google Cloud grant for deployment and scalability.
- Led the team to launch and maintain the platform serving 500+ monthly users from Victoria University of Wellington and the University of Canterbury.
- Designed and deployed a custom LMS plugin with retrieval-augmented generation for seamless integration into institutional systems.
- Configured secure database infrastructure and high-availability hosting, managing end-to-end development from concept through continuous user feedback.

Projects

Reaction Timer — Personal Project

Mar 2025

- Designed a high-precision reaction timer in VHDL on a Nexys-4 DDR FPGA board using an FSM architecture; integrated LEDs, seven-segment displays, and push buttons for user interaction and computed timing statistics via an ALU.

HaloVision — Motorcycle Safety HUD

2023 – 2024

- Open-sourced a helmet-mounted HUD integrating navigation and safety indicators on power-efficient custom PCBs.
- Designed multiple helmet mounts; used CFD and a closed-loop wind tunnel to verify compliance with NZ safety constraints.
- Developed a mobile app that pairs directly with the HUD for hands-free turn-by-turn navigation.

STM32 Step Counter — Personal Project

Apr 2024

- Developed STM32 step counter firmware with IMU-based detection and OLED display, featuring a fully reconfigurable interface. Allowing users to adjust step goals, measurement units, alert thresholds, and operational modes for dynamic customization.

Technical Skills

Languages: C, Python, VHDL, JavaScript

Tools & Frameworks: Vivado, Vitis, Next.js, Supabase, Docker, Google Cloud Platform

Other: CI/CD, Grafana, Git/GitHub, SolidWorks, KiCad

Extracurricular

Enduro mountain bike racing: Achieved 1st-place U21 finishes in regional NZ competitions.

Long-distance running: Completed multiple half-marathons and 20K trail runs.

Study group leadership: Organized and run large, peer-led weekly study sessions at UC.